

Technical Analysis Guide

Complete analysis plan — original E1-S3 + new analyses for presentation requirements

Business idea: Community Hub Marketplace — a single-location hub combining grocery (sourced from local farms), café, telehealth + health screening + Rx coordination, vendor marketplace, and community services (SNAP/WIC/insurance enrollment/banking). Creates 15-30 local jobs.

Analysis structure: The original E1-S3 analyses remain — they pull all the data. Five new analyses (AG, BM, IP, PO, MS) are added to cover what the presentation requires that E1-S3 doesn't: agriculture, benchmarking, implementation feasibility, predicted outcomes, and success metrics.

Analysis overview

Analysis	What it covers	Presentation slide it feeds
E1: Poverty & income	Poverty rate, income levels, gender gap, age distribution of poverty	3 — Community Overview 5 — Key Findings
E2: Employment & worker exodus	Labor force, OnTheMap flows, commute times, transportation modes	3 — Community Overview 5 — Key Findings
E3: Employer insurance gap	Insurance by employment, industry structure, uninsured by age	3 — Community Overview 5 — Key Findings
H1: Disease burden	CDC PLACES: chronic disease, disability, social determinants, prevention paradox	5 — Key Findings
H2: Federal designations & provider counts	HPSA scores, MUA, CHR provider counts, preventable hospitalizations, life expectancy	4 — Benchmarking 5 — Key Findings
H3: Access barriers	4 barriers: physical, financial, digital, provider — cross-referenced from all files	5 — Key Findings
S1: Industry gaps	Missing NAICS sectors, missing healthcare subcategories, what exists	5 — Key Findings 6 — Proposed Solutions
S2: Worker flow & business case	Job deficit, commute cost, healthcare sector opportunity	5 — Key Findings 6 — Proposed Solutions
S3: 6 recommendations → Hub components	Each Hub component mapped to NAICS gap, CDC need, Census viability, HPSA funding	6 — Proposed Solutions
NEW — AG: Agriculture	Census of Ag: farms, sales, food paradox, producer demographics, local supply chain	3 — Community Overview 5 — Key Findings 6 — Proposed Solutions
NEW — BM: Benchmarking	IGS position vs other tracts, CHR 3-level for all measures, rural peer comparison	4 — Benchmarking
NEW — IP: Implementation feasibility	Stakeholders, funding programs + dollar amounts, cost estimates, timeline, risks	7 — Implementation Plan
NEW — PO: Predicted outcomes	Which IGS metrics the Hub moves, by how much, forecasted IGS score improvement	8 — Predicted Outcomes

NEW — MS: Metrics for success	Baselines from CDC/CHR/Census, target values, timeframes, tracking method	9 — Metrics for Success
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EXISTING ANALYSES: E1 through S3

These remain unchanged from the previous technical guide. Summary below for reference — full specs (exact filters, column names, expected values) are in the CLAUDE.md.

E1: Poverty depth and income

[census_tract_951100_data.csv](#) – Categories: Poverty, Poverty by Age, Income

WHY: Poverty is the first link in the causal chain. 33.7% poverty (3x US) with falling incomes (-9.8%) establishes that this community can't afford healthy food, which causes the chronic disease you'll show in H1. Women 65-74 are 3.3x more likely impoverished — the group most needing healthcare.

KEY VALUES: Poverty rate 33.7%. Per capita \$22,536. Median HH \$45,618 vs LA \$58,273 vs US ~\$75k. Male 25-34: 247 in poverty. Female 55-64: 113. Female 65-74: 60 vs Male 18.

CHARTS: Poverty by age/sex grouped bar. Income 3-level comparison.

E2: Employment collapse and worker exodus

[census_tract_951100_data.csv](#) (Employment, Transportation, Travel Time) + [otm_inandout.pdf](#)

WHY: Shows physical isolation. 90.4% of workers leave daily, zero public transit, 38.2% commute 60+ min. This directly explains why people can't reach the appointment-only hospital. For the Hub: proves that ONE local location eliminates the commute barrier.

KEY VALUES: LFP 50.9%. Not in LF 49.1%. Unemployment 26.1%. OnTheMap: 2,880 leave, 305 local, 924 commute in. Deficit 1,956. Public transit: 0. 60-89 min bracket: 339 (largest).

CHARTS: C2 (stacked employment), C4 (travel time histogram), C10 (worker exodus infographic).

E3: Employer insurance gap

[census_tract_951100_data.csv](#) – Categories: Insurance by Employment, Industry, Health Insurance

WHY: The bombshell. Only 7.9% of employed workers have employer insurance (US ~55%). Destroys the assumption that jobs = healthcare. Women dominate food/retail (no benefits). For the Hub: the Health Zone accepts Medicaid (43.7% of parish) and doesn't depend on employer coverage. Community Services helps enroll the uninsured.

KEY VALUES: Employer-based: 38/483 = 7.9%. No insurance: 91 = 18.8%. Medicaid: 37 = 7.7%. Uninsured 35-64: 247 (86% of uninsured). Female Arts/Food: 214. Female Retail: 187. Zero women in Education/Health or Professional/Mgmt.

CHARTS: C3 (7.9% vs 27.6% vs 55% — THE #1 VISUAL), C5 (industry by gender).

H1: Disease burden

[CountiesCompareData_03142026.csv](#) (encoding utf-8-sig) + [CHR XLSX](#)

WHY: Proves poverty from E1 produces epidemic disease through food insecurity. The CDC data shows every condition above the US rate — and the Prevention category proves residents ARE trying (checkups 83.4% = above US) but dental collapses because zero dentists exist. For the Hub: the grocery addresses food insecurity → obesity → diabetes chain. The Health Zone catches HBP/diabetes through screening. The prevention paradox proves the Hub will be USED.

KEY VALUES: HBP 49.3 vs 34.4. Obesity 42.7 vs 32.8 (CHR: 46.3 vs LA 40.3). Diabetes 19.8 vs 12.0. Disability 45.6 vs 30.0. Mobility 24.3 vs 13.5. Food insecurity 31.7 vs 14.7. Checkups 83.4 vs 77.7. Dental 50.8 vs 63.9 (-13). Life expectancy: 70 vs LA 73.3 vs US 78.

CHARTS: C1 (8 chronic diseases), C6 (social determinants), C9 (disability), C13 (life expectancy).

H2: Federal designations and provider counts

[Hpsa_Find_Export.xlsx](#) + [Data_Explorer_Dataset.xlsx](#) + [CHR XLSX](#)

WHY: Federal government confirms the system is broken — 4 designations. CHR gives visceral provider counts: 2 PCPs, 1 dentist for 10,774. Preventable hospitalizations prove the shortage has consequences. For the Hub: HPSA/MUA designations unlock federal funding (NHSC recruitment, Section 330 grants, loan repayment).

KEY VALUES: HPSA: MH 20/25, Dental 22/26, PC 15/25, MCTA 22/25. FTE gap: 2.64. MUA: IMU 52.5. CHR: 2 PCPs, 1 dentist, 9 MH. Preventable hosp: 4,862 vs LA 3,427 vs US 2,666.

CHARTS: C11 (provider counts), C12 (preventable hosp 3-level).

H3: Access barriers | S1: Industry gaps | S2: Worker flow | S3: Recommendations

H3 cross-references all files into 4 barriers (physical, financial, digital, provider). The Hub solves all 4 simultaneously — ONE location (physical), Medicaid/SNAP (financial), on-site broadband (digital), telehealth to external providers (provider).

S1 uses CBP to show 10/20 sectors missing, 12/16 healthcare NAICS = 0. Each Hub component fills a documented NAICS gap.

S2 calculates job deficit (1,956), commute cost (\$4.37M/yr), and proves healthcare is already the #1 sector (450 emp, \$15.7M) — you're filling holes, not starting from nothing.

S3 maps each Hub component to its NAICS gap + CDC need + Census viability + HPSA funding.

CHARTS: C7 (sector heatmap), C8 (healthcare gaps table).

NEW ANALYSES: What the presentation requires that E1-S3 doesn't cover

AG: Agriculture analysis — the food paradox

[Census of Agriculture county profile \(cp22091.pdf\)](#)

WHY WE NEED THIS: The CBP data shows NAICS 11 (Agriculture) = 0 establishments, but that's misleading because CBP excludes farms. The Census of Agriculture reveals 349 farms producing \$54.2M in sales. This creates the most powerful finding for the Hub's grocery component: the parish PRODUCES food but residents face a food desert. The food leaves. The Hub keeps it local.

WHAT WE EXPECT TO FIND: A productive agricultural economy that doesn't serve its own community — proving the grocery component has a ready supply chain.

What to pull	Value	Why it matters
Number of farms	349 farms. Average 150 acres. 52,455 acres in farmland.	Agriculture is real and active — 349 operations ready to supply the Hub grocery.
Total sales	\$54.2M (+164% since 2017). Ranked 34th of 64 LA parishes. Net cash income: \$6.58M (+316%).	Agriculture is GROWING. Revenue up 164%. This is a thriving sector that could supply a local grocery.
What they produce	96% livestock/poultry. Poultry/eggs: \$43.7M (7th in LA). Cattle: \$4.3M. Dairy: \$4.1M (3rd in LA). 2.2M egg-laying hens. Crops: only \$1.98M (4%).	Eggs, dairy, beef are produced locally. Vegetables/fruits: \$271k + \$110k. Small but Hub demand could grow local produce farming.
The food paradox	\$54.2M produced locally BUT: CDC: food insecurity 31.7%. USDA: extreme food desert. CHR: Food Env Index 6.7/10. CDC: food stamps 29.5%.	THIS IS YOUR STRONGEST GROCERY ARGUMENT. The food exists. It just leaves. The Hub keeps it in the community.
Farm structure	97% family farms. Only 13% hire labor. 246 new/beginning farmers (41%). 73% have internet.	Explains why NAICS 11 = 0 in CBP and why employer insurance is 7.9%. Family farms = self-employed = no benefits. Hub provides health access for farmers too.
Producer demographics	606 producers. Female: 233 (38%). Black: 97 (16%). Age 65+: 254 (42%). 5% sell directly to consumers.	Only 5% sell direct to consumers. Hub vendor marketplace increases this. 42% of producers are 65+ = aging workforce needing health access.

CHART: C14 — Food paradox visual: \$54M produced locally vs 31.7% food insecure vs USDA food desert.

BM: Benchmarking — positioning St. Helena in context

[CHR Louisiana XLSX \(FIPS 22091 vs 22000\)](#) + [CHR National XLSX \(Sources sheet\)](#) + [IGS Report](#)

WHY WE NEED THIS: Slide 4 specifically requires 'IGS ranking compared to state and national averages.' The E1-S3 analyses compare individual metrics but never step back to position the community as a whole. This analysis shows St. Helena is worse than Louisiana on EVERY measure, and Louisiana is already near the bottom nationally.

WHAT TO DO:

Benchmarking task	How to do it	What you expect to find
IGS pillar comparison	From IGS report: Place 44, Economy 30, Community 33, Overall 36. Benchmarks are in the IGS report for each metric (column showing benchmark).	Every metric below benchmark except Affordable Housing (99), Early Ed (97), Net Occupancy (81). Show the 3 strong vs 15 weak.
CHR 3-level for every key measure	Pull FIPS 22091 (St. Helena) and FIPS 22000 (Louisiana state) from LA file. Pull US values from Sources sheet ("US Overall" column) of national file. Compare all three for each measure.	St. Helena worse than LA on every measure. LA worse than US on most. St. Helena = bottom of a state that's already near bottom.
Rural peer comparison (optional but strong)	From national CHR file, filter counties with % Rural > 90%, Population < 15,000, and Median HH Income < \$55,000. Compare St. Helena's health measures to this rural peer average.	Shows whether St. Helena is bad even by RURAL standards — or if this is "just what rural looks like." If worse than rural peers, argument is much stronger.
Preventable hosp 3-level	St. Helena: 4,862 Louisiana: 3,427 US: 2,666 Top performers: 1,596	St. Helena is 1.8x the US rate and 1.4x the state rate. Even by Louisiana standards — which are already bad — this is extreme.

CHART: C13 (life expectancy 3-level), C12 (preventable hosp 3-level), plus a new chart showing multiple CHR measures all benchmarked St. Helena vs LA vs US on one visual.

IP: Implementation feasibility

[HPSA/MUA XLSX + web research + Census of Agriculture + all existing data](#)

WHY WE NEED THIS: Slide 7 requires 'stakeholders, execution steps, risks, assumptions, constraints.' E1-S3 proved the problem and justified the Hub. This analysis asks: is it actually doable? Who runs it, who funds it, what does it cost, and what could go wrong?

WHAT TO ANALYZE:

Element	Data to research / pull	What to produce
Stakeholders	HPSA XLSX → SCHS (existing FQHC, 27+ sites, could operate Health Zone as satellite). Web research → St. Helena Parish Hospital (25 beds, Critical Access). Census of Ag → LSU AgCenter extension office. 349 farms for grocery supply chain.	Stakeholder map: SCHS = health operator Local farms = grocery supply Parish govt = land/permits USDA = funding Federal = HPSA/MUA grants
Funding sources (with dollar amounts)	HPSA designations → NHSC loan repayment (up to \$50k for 2-year commitment in HPSA). MUA → Section 330 FQHC New Access Point grants (\$650k-\$1M/year). USDA → Rural Business Development Grants (up to \$500k). Community Facilities Direct Loan (up to \$10M+). FCC → Rural Digital Opportunity Fund. USDA ReConnect (up to \$25M for broadband).	Funding table: Program Amount What it covers Eligibility met. Total potential funding from all sources.
Cost estimates	Web research → rural grocery store startup costs (\$250k-\$500k). Telehealth booth setup (\$15k-\$30k per booth). Commercial kitchen (\$50k-\$150k). Building lease/construction in rural LA (\$10-\$20/sqft).	Rough cost model: Component Startup cost Annual operating cost. Compare to available funding.
Revenue model	Hub concept doc → 6 revenue streams: grocery sales, café, vendor rent, health fees, institutional contracts, delivery. Census: 10,774 residents × avg spend. Ag Census: institutional supply to schools.	Revenue projection: Stream Monthly estimate. Breakeven analysis.
Risks and mitigations	Broadband risk → mitigated by on-site connection. Poverty limits revenue → mitigated by SNAP/EBT acceptance + institutional contracts. Farm supply willingness → supported by 349 farms, only 5% sell direct (98% untapped). Workforce availability → 49.1% not in LF.	Risk matrix: Risk Likelihood Impact Mitigation backed by data.
Timeline	Research typical FQHC expansion timelines (12-18 months for New Access Point grants). USDA grant cycles. Construction in rural areas.	Phased timeline: Year 1: funding + build. Year 2: launch + operate. Year 3: expand + measure.

PO: Predicted outcomes — forecasted IGS improvement

[IGS Report + all existing data](#)

WHY WE NEED THIS: Slide 8 specifically asks for 'forecasted improvement in Inclusive Growth Score.' You need to show which of the 18 IGS metrics the Hub would move, estimate by how much, and project a new overall IGS score.

WHAT TO DO: Map each Hub component to the IGS metrics it would improve. Estimate a realistic score increase based on what the Hub changes. Be conservative — judges prefer realistic projections over optimistic ones.

IGS metric	Current value	Current score	Hub component that moves it	Realistic new score	Reasoning
Internet access	66.1%	3	Hub provides on-site broadband. Doesn't change home access.	3-10	Small bump — Hub adds access point but doesn't wire homes.
Travel time to work	33.0% under 35	1	15-30 Hub jobs are local = no commute for those workers.	5-15	15-30 of ~1,200 workers is small % but moves the needle from 1.
Labor market engagement	1.0	1	15-30 direct jobs + vendor opportunities + farm supply contracts.	5-15	Meaningful increase in labor participation from rock bottom.
Commercial diversity	17.7%	34	Hub adds grocery, café, health, vendor, services = multiple new NAICS.	40-50	Several new NAICS codes activated is a real structural change.
Minority/women businesses	1.8%	22	Vendor marketplace prioritizes minority/ women entrepreneurs.	25-35	Modest increase from vendor stalls. Depends on uptake.
New businesses	9.8%	43	Hub itself + vendors = new businesses created.	45-55	Hub + 5-10 vendor businesses is real growth.
Health insurance coverage	92.4%	39	Community Services helps enroll uninsured on-site.	42-48	687 uninsured in parish. Enrolling even 100-200 moves the rate.
Female above poverty	66.1%	12	Jobs in food/retail/ health where women already work.	15-20	15-30 jobs, many going to women in food/retail, modest poverty reduction.

HOW TO PROJECT NEW IGS: The IGS is calculated from 18 metrics across 3 pillars. Most metrics the Hub doesn't touch (Gini, Park Land, Residential RE, etc.) stay the same. The 8 metrics above improve modestly. Recalculate pillar averages with new scores, then overall IGS. A conservative estimate: IGS could move from 36 to approximately 40-44 — approaching the 45 threshold. The Hub alone won't cross 45 but it closes the gap meaningfully.

CHART: C16 — IGS radar/comparison: current 36 vs projected ~40-44, showing which metrics moved.

MS: Metrics for success — baselines, targets, tracking

CDC PLACES + CHR + Census ACS + Census of Agriculture – all used as BASELINES

WHY WE NEED THIS: Slide 9 requires 'indicators used to track long-term impact.' The analyses above gave us baselines for every metric. This analysis turns those baselines into a measurement framework: what we track, what the starting number is, what target we aim for, and by when.

What to track	Baseline (current data)	Target (3-year)	Data source for tracking	Hub component measured
Food insecurity rate	31.7% (CDC PLACES)	Below 25%	CDC PLACES (updated annually)	Grocery: local food access
Preventable hosp rate	4,862/100k (CHR 2025)	Below 4,000	CHR (updated annually)	Health Zone: screening + telehealth

Dental visit rate	50.8% (CDC PLACES)	Above 58%	CDC PLACES	Health Zone: dental referral + rotating clinic
Obesity rate	42.7-46.3% (CDC/CHR)	Below 40%	CDC PLACES / CHR	Grocery: healthy food Café: healthy meals
Employer insurance or any coverage	7.9% employer 92.4% any coverage	95%+ any coverage	Census ACS (annual updates)	Community Services: enrollment assistance
# local jobs	1,229 (OnTheMap)	1,259+ (+30 from Hub)	OnTheMap / QWI (annual)	All Hub components: 15-30 jobs
SNAP enrollment / food stamp usage	29.5% (CDC PLACES)	Increase (more accessing benefits)	CDC PLACES	Community Services: on-site enrollment
IGS Score	36	40-44	IGS (next update)	All components combined
Local farm-to- consumer sales	5% of farms sell direct (Ag Census)	15%+	Census of Ag (every 5 years)	Grocery + vendor marketplace: local sourcing

KEY PRINCIPLE: Every metric has a current baseline from a named dataset, a realistic target, a tracking source, and a connection to a specific Hub component. Judges can see exactly how you'd measure whether the Hub is working.

Complete chart list

#	Type	Data source	Title	Slide
C1	Horiz grouped bar	hADC Health Outcomes, 8 gaps	Chronic disease: St. Helena vs US	5
C2	Stacked bar	Census Employment, 3 segments	Employment: 62.4% not working	3
C3	Bar (3 bars)	Census Ins by Emp: 7.9/27.6/55%	Only 7.9% have employer insurance	3, 5
C4	Histogram	Census Travel Time, color-coded	Commute: 38.2% travel 60+ min	3
C5	Grouped bar	Census Industry, male/female	Employment by industry and gender	3
C6	Horiz grouped bar	hADC Social Needs, 7 + multipliers	Social determinants: St. Helena vs US	5
C7	Grid/heatmap	CBP 20 sectors, present vs missing	Half the economy doesn't exist	5
C8	Table	CBP 12 missing healthcare NAICS	Healthcare gaps: 12 NAICS = 0	5, 6
C9	Horiz grouped bar	hADC Disability, 7 measures	Disability: St. Helena vs US	5
C10	Infographic	OnTheMap: 2,880 / 305 / 924	Worker exodus: 90.4% leave daily	3
C11	3-level bar	CHR: 2 PCPs, 1 dentist, vs benchmark	Provider counts: 2 doctors, 1 dentist	4, 5
C12	3-level bar	CHR: 4,862 vs LA 3,427 vs US 2,666	Preventable hospitalizations	4, 5
C13	3-level bar	CHR: 70 vs LA 73.3 vs US ~78	Life expectancy gap: 8 years	4
C14	Comparison vis	Ag Census \$54M vs CDC 31.7% insecure food	Food paradox: produces food, residents go hungry	3, 5, 6
C15	Multi-measure 3-level chart	CHR: all key measures, St. Helena vs LA vs US	Benchmarking dashboard	4
C16	Radar or before/after comparison	IGS: current 36 vs projected 40-44	Forecasted IGS improvement	8
C17	Metrics dashboard	hADC/CHR/Census baselines + targets + timeframes	Success metrics tracking framework	9

Order of operations: Run E1 → E2 → E3 → H1 → H2 → H3 → S1 → S2 → S3 first (they pull all data and produce findings). Then run AG (agriculture). Then BM (benchmarking — needs all data already pulled). Then IP (implementation — needs findings + web research). Then PO (predicted outcomes — needs everything). Then MS (metrics — needs baselines from everything). Once all analyses are done, arrange into the 10-slide presentation.

Files: census_tract_951100_data.csv | CountiesCompareData_03142026.csv | cbp_st_helena_parish.csv | Hpsa_Find_Export.xlsx | Data_Explorer_Dataset.xlsx | otm_inandout.pdf | CHR Louisiana + National XLSX | Census of Agriculture cp22091.pdf | IGS Reports | qwi SVGs